

ALEJANDRO PERRET-GENTIL

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PROFILE

Physics B.Sc. candidate with industry experience in PCB design, cryogenic hardware, prototype integration, and system-level testing. Hands-on background in experimental instrumentation, sensors, scientific Python, and multidisciplinary product development.

ENGINEERING EXPERIENCE

Working Student - PCB Design | kiutra GmbH

Feb 2025 - Jun 2025

Munich, Germany (Remote)

- Designed multi-layer PCB layouts in KiCad for cryogenic measurement electronics operating at sub-Kelvin temperatures.
- Collaborated with hardware engineers to refine schematics and layouts and support system-level integration of instrumentation platforms.

Engineering Intern | kiutra GmbH

Jun 2024 - Aug 2024

Munich, Germany

- Produced PCB layouts in KiCad and assembled cryostat components including thermal buses, vacuum hardware, and low-noise wiring.
- Integrated thin-film components, capacitors, and cryogenic measurement pucks; supported prototype validation, troubleshooting, and system-level testing.

EDUCATION

B.Sc. Physics | Constructor University Bremen

Sep 2022 - Present

Minor in Global Economics and Management | Bremen, Germany

- Relevant coursework: Advanced Physics Lab I & II, Robotics Lab, Numerical Software Lab, Analytical Mechanics, Electrodynamics, Statistical Physics, and Quantum Mechanics.

SELECTED TECHNICAL PROJECTS

GlucoCare - Low-Cost Non-Invasive Diabetes Monitoring Device

2024

YouthTeamUp x WaziUp | 13-person interdisciplinary team

- Researched and compared optical, thermal, and pulse-oximetry sensor options for a low-cost non-invasive glucose-monitoring concept, balancing measurement capability, component cost, and feasibility for low-resource deployment.
- Contributed to and peer-reviewed Arduino/Python/GitHub technical documentation within a 13-person interdisciplinary team.

Experimental Physics & Instrumentation

Spring 2023 - Spring 2025

- Advanced Physics Lab I (Fall 2023): operated a scanning electron microscope, adjusting emission current, focus, and stigmation; characterized an electron beam using Helmholtz coils and evaluated measurement uncertainty.
- Modern Physics Lab (Spring 2023) and Advanced Physics Lab II (Spring 2025): used CASSY DAQ, magnetic/force/displacement sensors, multimeters, power supplies, optical benches, lasers, and photodetection; performed calibration, repeatability checks, data analysis, and troubleshooting.

Bachelor Thesis - Computational Statistical Physics

2026

- Implemented Monte Carlo simulation and random-matrix-theory analysis of the 2D Ising model in Python (NumPy, SciPy, Matplotlib).

CERTIFICATIONS

Fundamentals of Quantitative Modeling - University of Pennsylvania | **Product Development using AutoCAD** | **Google AI Professional Certificate** (AI-assisted data analysis, research & insights)

TECHNICAL SKILLS

Hardware & Product: KiCad PCB layout, soldering, prototyping, cryogenic & vacuum hardware, low-noise wiring, sensors, Arduino; AutoCAD (certification/coursework)

Test & Instrumentation: CASSY Lab, calibration, uncertainty/error propagation, prototype validation, experimental test setups, troubleshooting, optical & electromagnetic measurement

Programming & Data: Python, NumPy, SciPy, Matplotlib, GitHub; numerical methods, Monte Carlo simulation, curve fitting, optimization, FFT, statistical analysis, data visualization

Languages: Spanish (Native), English (C1+), German (A2)

LEADERSHIP & ACTIVITIES

CASSI Board of Directors - Student Representative (2019-2022): organized 15+ charity events and served as student-staff liaison; 60+ volunteer hours across 14 service events.

Competitive kitesurfing and mountain biking (10+ years)